



HT34X – Jacketed Vessel With Coil And Stirrer

The characteristics of the heat transfer using an external jacket or internal coil can be demonstrated together with the effect of stirring the vessel contents.

PRODUCT CODE: HT34X

Categories: Educational, HT Series - Computer Controlled Heat Exchange / Heat Transfer, HT30X Service Unit Heat Exchange Accessories

Tags: Armbus , Computer Controlled , Fluid Streams , Heat Transfer , HT34X , Jacketed Vessel With Coil & Stirrer , PID Control , Stirring

DESCRIPTION

Vessel Heating or cooling of a process liquid in a tank, either batchwise or with continuous product feed, is common practice throughout industry.

The characteristics of the heat transfer using an external jacket or internal coil can be demonstrated together with the effect of stirring the vessel contents.

TECHNICAL SPECIFICATIONS

The accessory consists of a jacketed vessel with the following features:

- The vessel consists of a stainless steel wall with PVC base and clear acrylic top. A glass outer jacket allows the wall of the vessel to be surrounded with hot fluid for indirect heating from the outside. Alternatively, a stainless steel coil inside the vessel allows the cold fluid contained within the vessel to be indirectly heated from inside.
- The vessel incorporates a variable speed stirrer and baffle arrangement to provide thorough mixing of the vessel contents when required.
- An adjustable overflow enables the volume of liquid inside the vessel to be varied with a maximum capacity of two litres and a minimum of one litre.
- The vessel can be operated batchwise by simply filling to the overflow or with continuous feed of cold liquid to the base of the vessel, the excess liquid flowing from the overflow to drain.
- Temperature can be measured at the following 5 locations:
 - Vessel contents (cold fluid)
 - Hot fluid inlet to jacket/coil
 - Hot fluid outlet from jacket/coil
 - Cold water outlet to drain
 - Cold fluid inlet to vessel

- Quick release hot and cold fluid connections enable rapid connection to HT30X and conversion from heating jacket to heating coil.

FEATURES & BENEFITS

- Miniature jacketed vessel heat exchanger system for use with an Computer Controlled Armfield Heat Exchanger Service Unit
- Comprises processing vessel with outer jacket, inner coil, variable speed stirrer and baffle
- Thermocouples measure the vessel contents and the inlet and outlet temperatures of both fluid streams (5 in total)
- The Heat Exchanger is designed to be installed on the service unit without the need for tools
- A comprehensive instruction manual is included

armSOFT Software Features

ArmBUS software is provided as standard with the HT30X for all the Armfield heat exchanger accessories. Individual programs are provided for each exchanger, and each program contains a selection of separate exercises that can be performed.

The actual details are exercise specific, but typically the following interfaces are available:

- Full Graphical User Interface (GUI) for each experiment with display screens tailored for each exercise
- All the temperatures and flow rates are displayed on a diagrammatic representation of the equipment
- Student questions and equations can be accessed individually for each exercise
- The hot-water flow rate utilises optional Manual/Automatic control to control the pump output. Manual control allows the user to select a percentage of the total power to supply to the pump. Automatic control allows the user to input the desired set point which the PID control will reach and maintain. The use of automatic control ensures the flow is stable despite changes in the viscosity of the water due to heating
- Real-time flow rate readings are displayed in L/min Sensor data is collated and calculations are displayed in a data-log, a tabulation function provided with the armBUS software. The data is in tabulated format and can be saved and accessed through a .csv file compatible with software such as Microsoft Excel
- The cold-water flow control valve utilises optional Manual/Automatic settings to control water flow from the mains. Manual control allows the user to open/close the valve between 0 and 100%. Automatic control allows the user to input the desired set point which the PID control will reach and maintain so long as the input pressure is adequate.
- The heater utilises optional Manual/Automatic control to supply power to the heater element in the boiler. Manual control allows the user to select a percentage of the total power to supply to the heater. Automatic control allows the user to select a set point (°C), and then PID control software heats the water to and maintains a set temperature
- The data from the sensors is plotted and displayed in a user-configurable graphing function of the software. It can be displayed separately from the data logging and each sensor output can be viewed independently
- Processing of measured values to obtain calculated results (this can be linked to the questions and answers to ensure student understanding, answers to equation calculations can be found as part of the datalogger output)

Essential Accessories

- **HT30X** Computer Controlled Heat Exchanger Service Unit

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.10m³

Gross weight: 9 kg

Dimensions



Length: 0.40m
Width: 0.18m
Height: 0.39m

Order Codes

HT34X – Jacketed Vessel With Coil & Stirrer

See HT30X for power options

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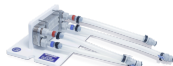
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